HONEST VALIDATION RESHAPES EEG-BASED PARKINSON'S DISEASE DETECTION

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INTRODUCTION

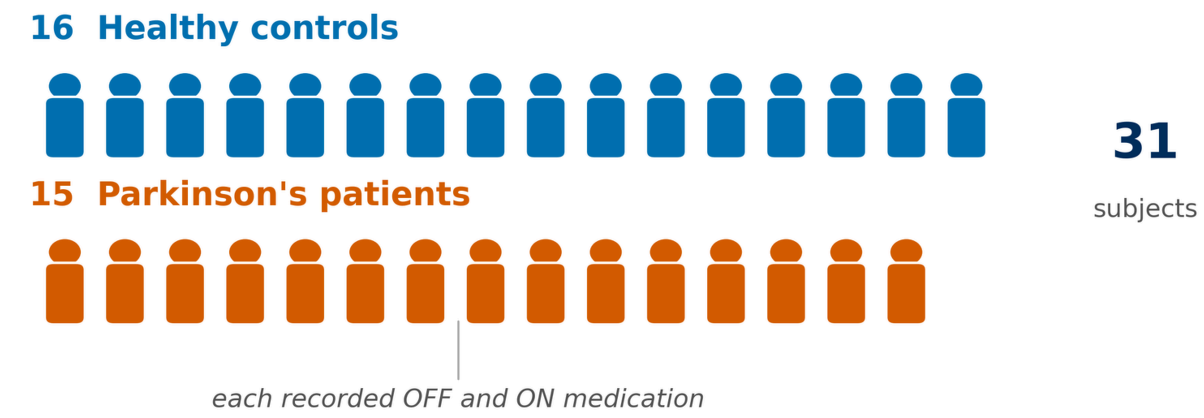
- Parkinson's disease (PD) is the second most common neurodegenerative disorder; its early clinical diagnosis remains highly subjective.
- Recent machine learning frameworks utilizing resting-state EEG report automated diagnostic accuracies nearing 100%.
- Most studies evaluate models segment-wise, meaning 10-second epochs from the same individual are mixed across both training and testing datasets.
- Rather than learning legitimate pathological biomarkers, models memorize individual anatomical signatures and idiosyncratic artifacts, drastically inflating performance numbers.

OBJECTIVE

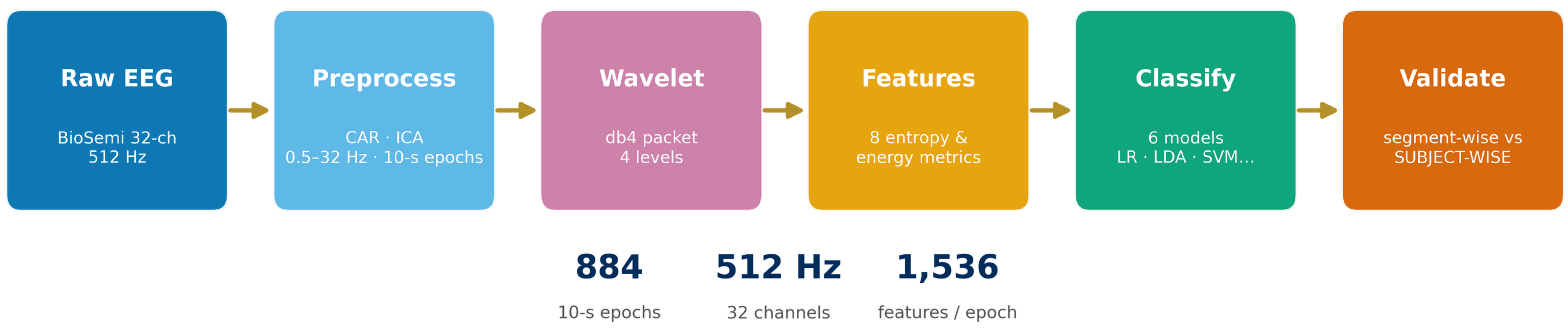
To systematically quantify the performance collapse between segment-wise and subject-wise validation by replicating a state-of-the-art framework, and to rigorously evaluate the feasibility of translating these models to accessible environments using the 14-channel configuration of the Emotiv Epoc consumer device.

METHODS

Dataset: UCSD resting-state EEG (Rockhill et al., 2021)

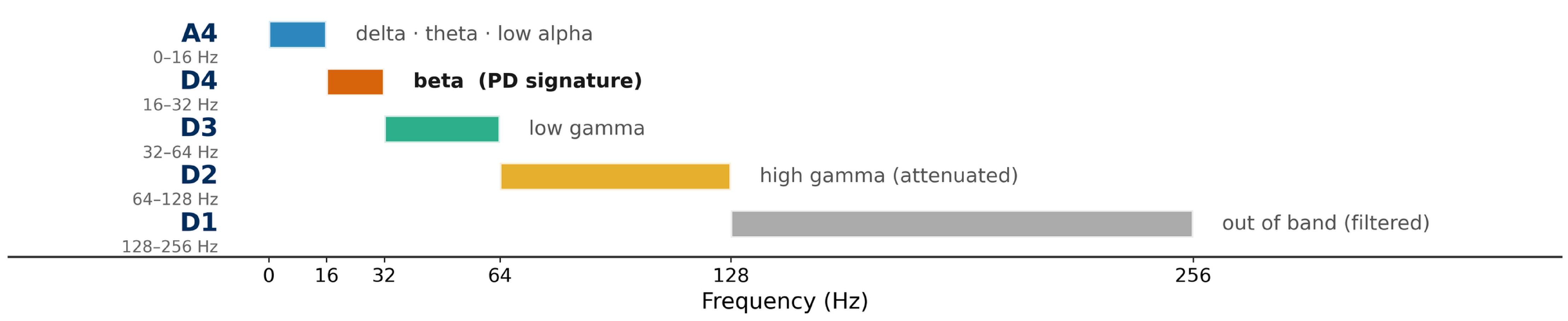


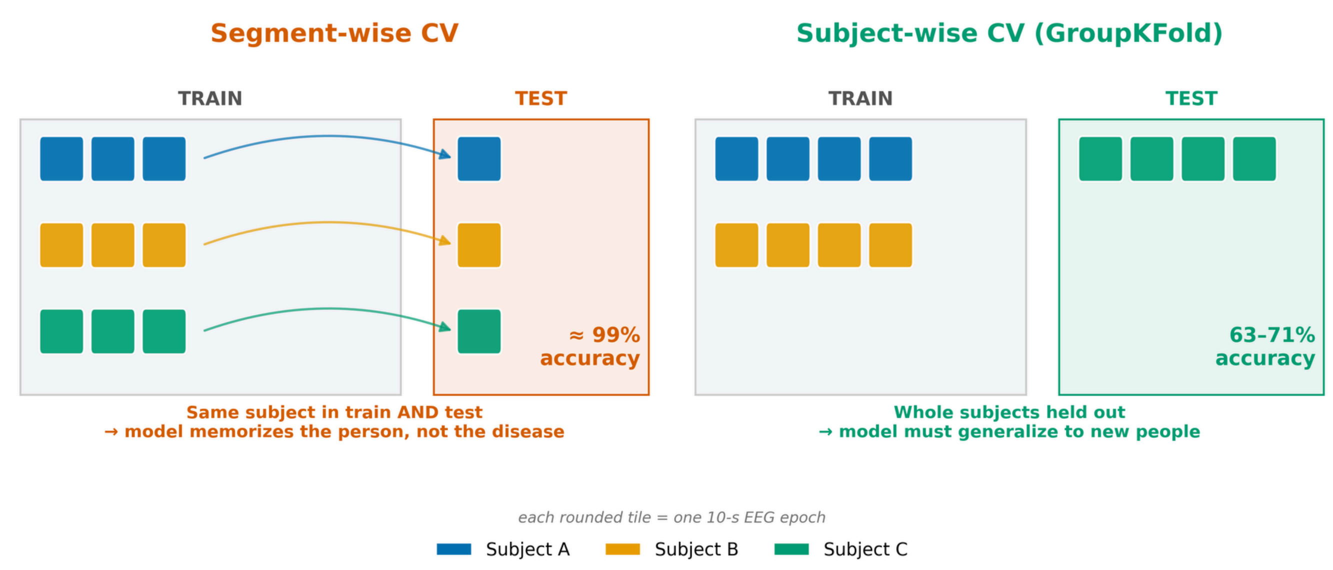
Processing pipeline (Aljalal et al., 2022, replicated)



- Public UC San Diego resting-state EEG dataset (n = 31; 15 PD patients in On/Off medication states, 16 healthy controls).
- Extended Infomax ICA + ICLabel automated classification to eliminate non-brain artifacts (exclusion threshold > 80% probability for eye, muscle, or line noise).
- 5th-order zero-phase Butterworth filtering (0.5–32 Hz) and non-overlapping slicing into 10-s epochs (884 total segments).
- Wavelet Packet Decomposition (db4, 4 levels) to extract time-frequency sub-bands **(focusing on the D4 detail coefficient: beta band, 16–32 Hz)**.
- Extraction of 8 distinct complexity metrics per channel: Energy, Local Band Power (LBP), Threshold, Norm, Sure, Log-Energy, Shannon, and T-Shannon entropy (Total feature dimension: 1,536).
- Benchmarking of 6 standard classifiers (LR, LDA, RF, SVM-linear, SVM-quadratic, KNN).
- Scheme A (Segment-wise): Standard 10×10 K-Fold cross-validation (allowing intra-subject data leakage).
- Scheme B (Subject-wise): 10 repetitions of 5-fold GroupKFold, strictly keeping whole subjects out of the training set.
- Statistical significance validated via label permutation testing (1,000 iterations).

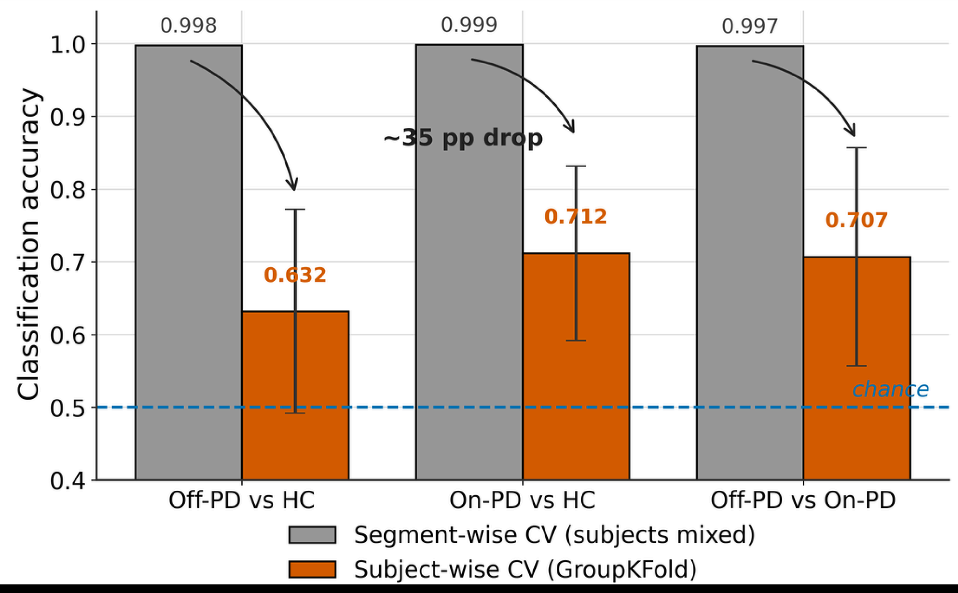
db4 wavelet packet decomposition (4 levels, fs = 512 Hz)





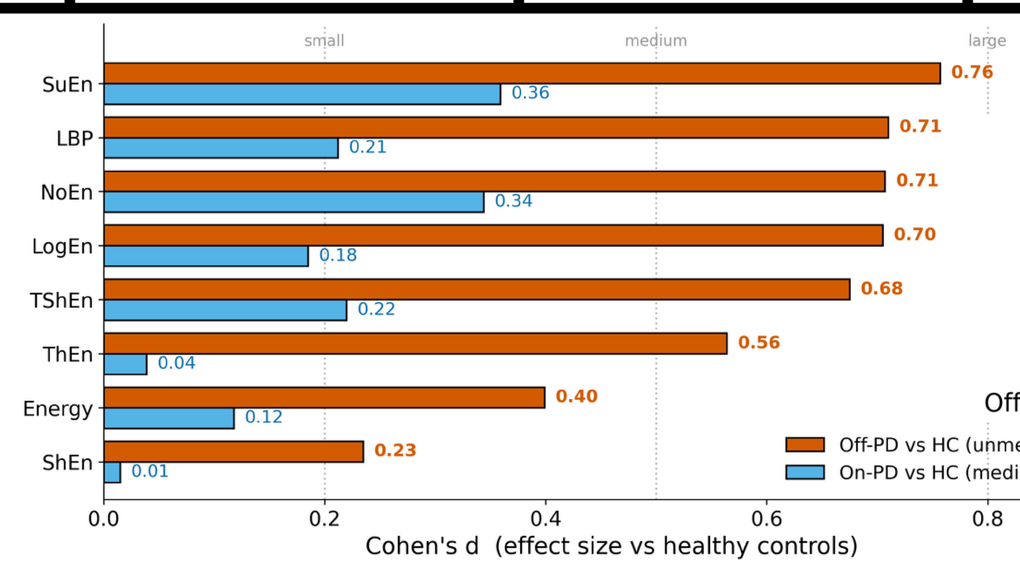
RESULTS

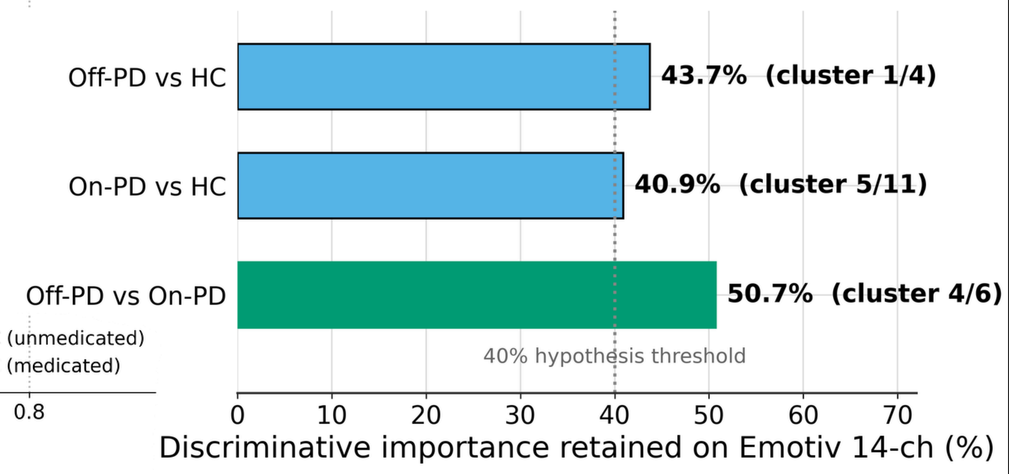
- Replicating the pipeline yielded a consistent **~35 percentage-point drop** in performance across all tasks when enforcing strict subject-wise isolation.
- Headline segment-wise accuracies (≥ 0.997) collapsed to a realistic range of 0.632–0.712 under honest evaluation.
- Crucially, all subject-wise models remained statistically superior to chance ($p = 0.001$), confirming a genuine underlying neural signal.

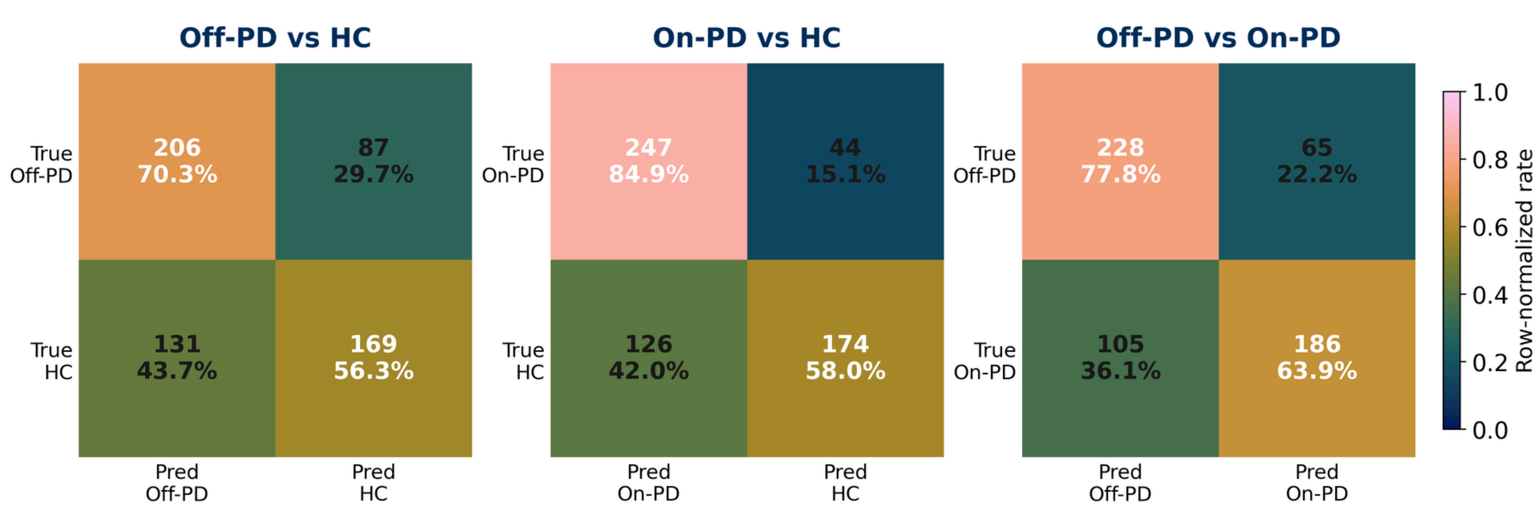


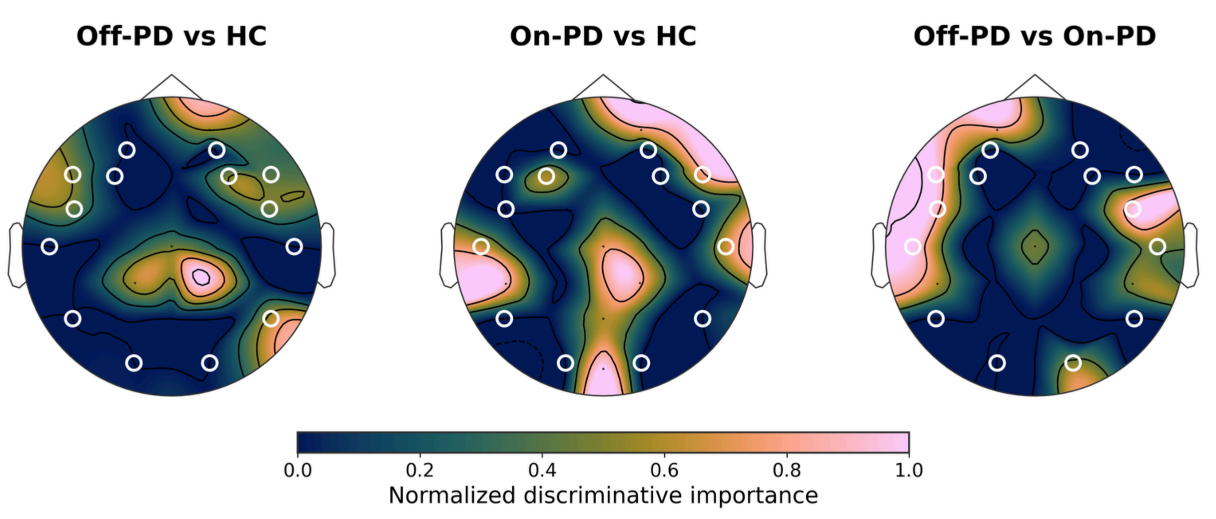
Clinical Classification Task	Optimal Model	Winning Feature	Accuracy	AUC
Off-PD vs HC	LDA	Norm Entropy (NoEn)	0.632 ± 0.14	0.628
On-PD vs HC	Logistic Reg.	Norm Entropy (NoEn)	0.712 ± 0.12	0.787
Off-PD vs On-PD	SVM-linear	Local Band Power (LBP)	0.707 ± 0.15	0.732

- The medication-state contrast (Off-PD vs On-PD) transferred best, retaining 50.7% of total discriminative importance and covering 4/6 channels within the significant cortical cluster (T7, F7, FC5, FC6).
- Conversely, direct diagnosis (Off-PD vs HC) relied heavily on the medial centro-parietal region (CP1, CP2), which is outside the spatial layout of the Emotiv headset (covering only 1/4 channels of the cluster).









CONCLUSIONS

1. Headline EEG-PD accuracies near 100% in published literature reflect subject-specific pattern memorization rather than true clinical generalization.
2. Honest subject-wise accuracy (63–71%) is modest but robust, statistically valid, and comparable to true inter-subject performance.
3. Monitoring dopaminergic medication states (On vs Off) is the most viable application for consumer-grade devices like Emotiv. However, hardware discrepancies (sampling rates, 14-bit quantization, and saline sensors) function as an optimistic upper bound that demands active empirical field testing.

FUTURE WORK

- Validate the pipeline across the larger UNM cohort (n = 54) to ensure topographical stability. Initiate an empirical pilot study using commercial headsets to build a local dataset, with plans to expand the methodology into Alzheimer's disease cohorts.

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REFERENCES